

Cryptography Master Exam Review (100 Questions)

Complete Ordered 100-Problem Question Bank with Verified Answers & Scannable Explanations

Total Questions: 100

Accuracy: 100% Verified

Standard: Hyphenated Fill-Ins & Core MCQs

1. $GF(2^8)$ arithmetic is directly used inside which widely deployed algorithm?

- A. The Euclidean Algorithm, when computing a greatest common divisor
- B. RSA, when selecting the two large prime factors of its modulus
- C. AES, in its byte substitution and column mixing steps
- D. The plain Caesar cipher, in choosing its shift amount

Correct Answer: C. AES, in its byte substitution and column mixing steps

Explanation: AES operates on 8-bit bytes defined as elements in $GF(2^8)$. SubBytes uses multiplicative inverses in $GF(2^8)$, and MixColumns performs polynomial matrix multiplication over $GF(2^8)$.

2. The theorem guaranteeing that every integer greater than 1 factors into primes in exactly one way is called the ____ Theorem of Arithmetic.

Correct Answer: fundamental-theorem

Explanation: The Fundamental Theorem of Arithmetic states that every integer greater than 1 has a unique prime factorization up to the order of factors.

3. The set of points on an elliptic curve, together with a defined addition rule and a point at infinity, forms an algebraic structure called an ____ group.

Correct Answer: abelian

Explanation: Elliptic curve point addition satisfies closure, associativity, identity, inverses, and commutativity, forming an abelian group.

4. The square-and-multiply technique for RSA is valuable because it computes a large modular exponentiation such as x^{16} using:

- A. Addition alone, avoiding multiplication entirely
- B. Repeated squaring, needing only a handful of multiplications instead of performing the full exponent's worth
- C. A random guess-and-check process repeated until the result matches
- D. A single lookup table indexed by the exponent value

Correct Answer: B. Repeated squaring, needing only a handful of multiplications instead of performing the full exponent's worth

Explanation: Square-and-multiply computes modular exponentiation in $O(\log e)$ operations using repeated squarings and conditional multiplications.

5. A base g whose successive powers modulo a prime p cycle through every nonzero remainder before repeating is called a ____ of p .

Correct Answer: primitive-root

Explanation: A primitive root modulo p generates all $p - 1$ non-zero elements in $\mathbb{Z}_p^*$.

6. AddRoundKey is the only AES transformation that:

- A. Requires computation in $GF(2^8)$ using a fixed multiplication matrix
- B. Provides nonlinearity through a substitution table
- C. Reorders bytes without changing any of their values
- D. Directly incorporates the secret key material into the State, via XOR with the round key

Correct Answer: D. Directly incorporates the secret key material into the State, via XOR with the round key

Explanation: SubBytes, ShiftRows, and MixColumns do not use key material; AddRoundKey is the only round operation that XORs the round key into the State.

7. For a positive integer n , arithmetic performed modulo n confines every result to the range:

- A. $\{0, 1, \dots, 2n-1\}$, doubling the usual modular range for security
- B. $\{1, 2, \dots, n\}$, excluding zero entirely from every calculation
- C. $\{-n, \dots, 0, \dots, n\}$, allowing both positive and negative remainders freely
- D. $\{0, 1, \dots, n-1\}$, wrapping around whenever a value would leave that range

Correct Answer: D. $\{0, 1, \dots, n-1\}$, wrapping around whenever a value would leave that range

Explanation: Arithmetic modulo n maps all results into the standard complete set of residues $\{0, 1, \dots, n-1\}$.

8. The attack that defeats Double DES by working forward from the plaintext and backward from the ciphertext to find a matching intermediate value is called the ____ attack.

Correct Answer: meet-in-the-middle

Explanation: Meet-in-the-middle matches forward encryption $E_{K1}(P)$ with backward decryption $D_{K2}(C)$ in an intermediate table, reducing effective security to $\sim 2^{57}$.

9. The problem of finding the exponent x that solves $g^x \equiv b \pmod{p}$, believed to be computationally infeasible for large p , is called the ____ problem.

Correct Answer: discrete-logarithm

Explanation: The Discrete Logarithm Problem (DLP) is the one-way hard mathematical problem underlying classical Diffie-Hellman and ElGamal.

10. The final round of AES encryption differs from the earlier rounds in that it:

- A. Omits the AddRoundKey transformation entirely
- B. Repeats SubBytes twice in succession
- C. Omits the MixColumns transformation
- D. Uses a different, smaller S-box than earlier rounds

Correct Answer: C. Omits the MixColumns transformation

Explanation: The final round consists only of SubBytes, ShiftRows, and AddRoundKey, omitting MixColumns to make decryption structurally symmetrical.

11. In RSA key generation, once primes p and q have been chosen, the public modulus is computed as $n = p \cdot q$ and the totient is computed as:

- A. $\phi(n) = p \cdot q - 1$
- B. $\phi(n) = (p-1)(q-1)$
- C. $\phi(n) = p + q - 1$
- D. $\phi(n) = (p-1) + (q-1)$

Correct Answer: B. $\phi(n) = (p-1)(q-1)$

Explanation: Euler's totient function for the product of two distinct primes p and q is $\phi(n) = \phi(p) \cdot \phi(q) = (p-1)(q-1)$.

12. The fast, widely used probabilistic primality test that repeatedly tests random witnesses against a candidate number is called the ____ test.

Correct Answer: Miller-Rabin

Explanation: The Miller-Rabin test checks whether candidate witnesses satisfy modular square root properties of primes.

13. Elliptic curve cryptography is attractive compared to RSA mainly because, for a comparable level of security, ECC:

- A. Requires substantially smaller key sizes, reducing processing overhead
- B. Has a longer and more thoroughly tested track record than RSA
- C. Does not require the two communicating parties to agree on any shared curve parameters
- D. Eliminates the need for any private key on either side of the exchange

Correct Answer: A. Requires substantially smaller key sizes, reducing processing overhead

Explanation: ECC provides equivalent cryptographic security with much smaller keys (e.g., 256-bit ECC matches 3072-bit RSA), saving compute and bandwidth.

14. In a stream cipher, the pseudorandom sequence of bits combined with the plaintext via XOR to produce ciphertext is called the ____.

Correct Answer: keystream

Explanation: Stream ciphers generate a pseudorandom keystream that is XORed bitwise with the plaintext.

15. In $GF(2^m)$, addition of two field elements is performed using the bitwise operation ____.

Correct Answer: XOR

Explanation: Polynomial addition over $GF(2)$ corresponds to coefficient addition modulo 2, which is equivalent to bitwise XOR.

16. Diffie and Hellman's 1976 breakthrough addressed which two problems inherent in purely symmetric cryptography?

- A. The need for a trusted certificate authority and the cost of hardware acceleration
- B. Resistance to brute-force attacks and resistance to differential cryptanalysis
- C. Encryption speed and the size of the ciphertext produced for short messages
- D. Secure key distribution without a pre-shared secret, and the lack of a scheme equivalent to a handwritten digital signature

Correct Answer: D. Secure key distribution without a pre-shared secret, and the lack of a scheme equivalent to a handwritten digital signature

Explanation: Public-key cryptography solved key exchange over insecure channels and created digital signature mechanisms.

17. Why is MixColumns skipped in the final AES round rather than in every round?

- A. Because skipping it doubles the effective key length of the cipher
- B. Because MixColumns cannot be computed on the very last block of a message
- C. To keep encryption invertible and simple while still ensuring the earlier rounds provide adequate diffusion
- D. Because the final round always uses a 256-bit key regardless of the chosen AES variant

Correct Answer: C. To keep encryption invertible and simple while still ensuring the earlier rounds provide adequate diffusion

Explanation: Skipping MixColumns in the last round makes the inverse cipher structurally identical to the forward cipher without compromising diffusion.

18. According to the design-parameter discussion of Feistel ciphers, below a certain number of rounds a cipher becomes more vulnerable because:

- A. The S-boxes lose their nonlinear behavior entirely
- B. Differential cryptanalysis can become cheaper than an exhaustive key search
- C. The block size effectively shrinks with each additional round
- D. The key schedule stops generating distinct subkeys for each round

Correct Answer: B. Differential cryptanalysis can become cheaper than an exhaustive key search

Explanation: Too few rounds leave statistical differentials and linear paths detectable, making cryptanalysis faster than brute force.

19. The Blum Blum Shub (BBS) generator produces each output bit by:

- A. Computing the discrete logarithm of the previous output modulo a large prime
- B. XORing the previous two output bits together, similar to a shift register
- C. Directly encrypting a counter value using AES in CTR mode
- D. Repeatedly squaring a value modulo $n = p \cdot q$ and taking the least significant bit of the result each round

Correct Answer: D. Repeatedly squaring a value modulo $n = p \cdot q$ and taking the least significant bit of the result each round

Explanation: BBS evaluates $x_{i+1} = x_i^2 \bmod n$ ($n = p \cdot q$) and outputs the least significant bit (or parity bit) at each step.

20. The AES State array arranges the 16 bytes of a block into a matrix of:

- A. 8 rows by 2 columns
- B. 2 rows by 8 columns
- C. 4 rows by 4 columns
- D. 16 rows by 1 column

Correct Answer: C. 4 rows by 4 columns

Explanation: The 16 bytes of a 128-bit block are arranged in a 4-row by 4-column column-major matrix.

21. The AES round transformation that performs a non-linear byte-by-byte substitution using a fixed lookup table is called ____.

Correct Answer: SubBytes

Explanation: SubBytes uses an S-box lookup based on $GF(2^8)$ inversion and an affine mapping to provide non-linear confusion.

22. A deterministic algorithm that takes a seed and produces a long sequence of numbers that statistically resemble true randomness is called a ____.

Correct Answer: pseudorandom-number-generator

Explanation: A Pseudorandom Number Generator (PRNG) deterministically expands a short random seed into a longer pseudorandom sequence.

23. A finite field, or Galois field $GF(q)$, is a finite set of elements with addition and multiplication such that:

- A. Every element, including zero, is required to have a multiplicative inverse
- B. Only addition is guaranteed to have inverses; multiplication need not be invertible
- C. The number of elements q must always be an even number
- D. Every nonzero element has a multiplicative inverse and the usual algebraic laws (associativity, commutativity, distributivity) hold

Correct Answer: D. Every nonzero element has a multiplicative inverse and the usual algebraic laws (associativity, commutativity, distributivity) hold

Explanation: A field is an algebraic structure where addition and multiplication are commutative and every non-zero element has a multiplicative inverse.

24. In an extension field $GF(2^m)$, each element can be represented as:

- A. A single decimal digit between 0 and m
- B. A polynomial of degree less than m with binary (0 or 1) coefficients, equivalent to an m -bit string
- C. An ordered pair of prime numbers whose product is less than 2^m
- D. A matrix of size m by m containing only prime entries

Correct Answer: B. A polynomial of degree less than m with binary (0 or 1) coefficients, equivalent to an m -bit string

Explanation: Elements of $GF(2^m)$ are polynomials of degree $\leq m - 1$ with binary coefficients $\{0, 1\}$.

25. The efficient method that finds the greatest common divisor of two integers through repeated division is called the ____ Algorithm.

Correct Answer: Euclidean

Explanation: The Euclidean Algorithm finds $\gcd(a, b)$ in logarithmic time by repeatedly taking division remainders.

26. The block cipher mode that XORs each plaintext block with the previous ciphertext block before encryption, requiring an initialization vector for the first block, is called ____ mode.

Correct Answer: Cipher-Block-Chaining

Explanation: Cipher Block Chaining (CBC) encrypts $C_i = E_K(P_i \text{ XOR } C_{i-1})$ with $C_0 = IV$.

27. RSA was developed in 1977 by Ron Rivest, Adi Shamir, and Len ____, whose initials give the algorithm its name.

Correct Answer: Adleman

Explanation: RSA is named after its creators: Ron Rivest, Adi Shamir, and Leonard Adleman.

28. Applying Fermat's Little Theorem with $a = 3$ and $p = 7$, the value of $3^6 \bmod 7$ must equal:

- A. 6
- B. 3
- C. 1
- D. 0

Correct Answer: C. 1

Explanation: Fermat's Little Theorem states that $a^{p-1} \equiv 1 \pmod p$ for prime p coprime to a . Here $3^6 \equiv 1 \pmod 7$.

29. The block cipher mode specifically designed for encrypting data on sector-based block storage devices (data-at-rest) is called ____.

Correct Answer: XTS-AES

Explanation: XTS-AES is the tweakable block cipher standard designed for sector-level storage encryption.

30. Feistel's design goal was to approximate the security of an ideal block cipher while avoiding its main practical drawback, which is that an ideal cipher with an n -bit block requires:

- A. A key large enough to select among $2^n!$ possible transformations, which is computationally unmanageable
- B. A separate physical key exchange for every message sent
- C. A trusted third party to certify the block size in advance
- D. Hardware capable of performing floating-point arithmetic

Correct Answer: A. A key large enough to select among $2^n!$ possible transformations, which is computationally unmanageable

Explanation: An ideal block cipher requires specifying an arbitrary permutation from $(2^n)!$ possibilities, requiring an impractical key length of $\sim n \cdot 2^n$ bits.

31. An element a is called a primitive root of a prime p when:

- A. a raised to any power always produces the same remainder modulo p
- B. The successive powers of a , taken modulo p , cycle through every nonzero remainder before repeating
- C. a is the smallest prime factor of p minus one
- D. a itself must also be prime and strictly greater than p

Correct Answer: B. The successive powers of a , taken modulo p , cycle through every nonzero remainder before repeating

Explanation: A primitive root generates the complete set of non-zero residues modulo p .

32. An integer p greater than 1 is defined as prime when:

- A. It cannot be expressed as the product of any two smaller integers at all
- B. Its only positive divisors are 1 and itself
- C. It has exactly three distinct positive divisors, including itself
- D. It is odd and also not divisible by 3 or 5

Correct Answer: B. Its only positive divisors are 1 and itself

Explanation: A prime number is an integer greater than 1 with exactly two distinct positive divisors: 1 and itself.

33. The value that provides variability so that identical plaintext blocks do not produce identical ciphertext at the start of CBC or CFB encryption is called the ____.

Correct Answer: initialization-vector

Explanation: The Initialization Vector (IV) provides uniqueness and randomness to prevent identical ciphertexts under the same key.

34. In a stream cipher, ciphertext is produced by:

- A. Substituting each plaintext block with a fixed-size hash of that block
- B. Passing the plaintext through several parallel S-boxes without any key material
- C. Multiplying the plaintext by the key modulo a large prime number
- D. XORing the plaintext with a keystream generated from the secret key by a PRNG

Correct Answer: D. XORing the plaintext with a keystream generated from the secret key by a PRNG

Explanation: Stream ciphers encrypt by XORing plaintext bits directly with pseudorandom keystream bits.

35. The statement $a \equiv b \pmod{n}$ means that:

- A. a multiplied by b is always evenly divisible by n
- B. n divides $(a - b)$, so a and b leave the same remainder when divided by n
- C. a and b are both prime numbers that happen to share the modulus n
- D. a and b must be equal integers before applying the modulus operation

Correct Answer: B. n divides $(a - b)$, so a and b leave the same remainder when divided by n

Explanation: $a = b \pmod{n}$ implies n divides $(a - b)$ without remainder.

36. XTS-AES ties encryption to a particular physical storage location by using:

- A. A running counter that resets to zero once per day
- B. A separate hash function computed over the entire disk
- C. A second, independently generated AES key for every sector
- D. A tweak value derived from the sector or block address, alongside the key

Correct Answer: D. A tweak value derived from the sector or block address, alongside the key

Explanation: XTS-AES encrypts the sector address as a tweak value to make ciphertext dependent on disk physical locations.

37. DES uses an initial permutation, ____ rounds of processing, and then the inverse of the initial permutation to produce the ciphertext.

Correct Answer: 16

Explanation: Standard DES executes 16 Feistel rounds between the Initial Permutation and Inverse Initial Permutation.

38. A message encrypted only with the sender's private key, as in a basic digital signature scheme, provides authentication but NOT confidentiality because:

- A. Digital signatures are only ever applied to a hash of the message, never its content
- B. Anyone who has the sender's public key can decrypt and read the message
- C. The message is never actually transformed, only tagged with a plaintext label
- D. The private key used is always shorter than the key used for encryption alone

Correct Answer: B. Anyone who has the sender's public key can decrypt and read the message

Explanation: Because the corresponding public key is openly available, any observer can decrypt and read the message.

39. The elliptic curve discrete logarithm problem, the hard problem underlying ECC security, asks an attacker to:

- A. Recover the integer k given a base point P and the point kP obtained by repeated point addition
- B. Factor the coefficients a and b that define the elliptic curve equation
- C. Compute the y -coordinate of a point given only its x -coordinate, without a curve equation
- D. Find two distinct points on the curve that sum to the point at infinity

Correct Answer: A. Recover the integer k given a base point P and the point kP obtained by repeated point addition

Explanation: ECDLP is the challenge of determining scalar integer k given base point P and scalar multiple $Q = kP$.

40. The recommended padding scheme that randomizes RSA plaintext before encryption, defending against chosen-ciphertext attacks, is called ____.

Correct Answer: OAEP

Explanation: Optimal Asymmetric Encryption Padding (OAEP) provides semantic security and chosen-ciphertext resistance for RSA.

41. Concern about the DES S-boxes containing a deliberate weakness arose mainly because:

- A. IBM and the NSA never publicly disclosed the exact criteria used to design them
- B. They were replaced entirely in the final published version of the standard
- C. They were later shown to reduce the effective key length to under 40 bits
- D. Independent researchers had mathematically proven a fatal structural flaw in them

Correct Answer: A. IBM and the NSA never publicly disclosed the exact criteria used to design them

Explanation: Classified S-box design criteria led to suspicions of trapdoors, though they were actually optimized against differential attacks.

42. Euler's Theorem, $a^{\phi(n)}$ is congruent to 1 (mod n) provided that $\gcd(a, n) = 1$, is best described as:

- A. An unrelated result used exclusively for testing whether n is a perfect square
- B. A generalization of Fermat's Little Theorem that applies to any modulus n , not only prime moduli
- C. A restatement of the Euclidean Algorithm using exponents instead of remainders
- D. A special case of Fermat's Little Theorem that only applies when n itself is prime

Correct Answer: B. A generalization of Fermat's Little Theorem that applies to any modulus n , not only prime moduli

Explanation: Euler's Totient Theorem generalizes Fermat's Little Theorem from prime moduli to arbitrary composite moduli.

43. Claude Shannon's principle by which each plaintext digit affects many digits of the ciphertext, hiding statistical structure, is called ____.

Correct Answer: diffusion

Explanation: Diffusion spreads single-bit plaintext differences across multiple ciphertext output bits.

44. RC4's real-world downfall in Wi-Fi security came primarily through WEP because:

- A. WEP replaced RC4 with a much weaker, custom-designed stream cipher
- B. RC4 cannot process variable-length keys, forcing WEP to use a single fixed key
- C. WEP's method of generating and reusing keys fed into RC4 was flawed, not the RC4 algorithm itself in isolation
- D. RC4 itself was mathematically broken and could be reversed without any key at all

Correct Answer: C. WEP's method of generating and reusing keys fed into RC4 was flawed, not the RC4 algorithm itself in isolation

Explanation: WEP concatenated 24-bit predictable IVs with the key, causing keystream reuse and enabling FMS key recovery.

45. A common misconception about public-key cryptography, according to the chapter, is the belief that it:

- A. Cannot be used for digital signatures, only for encrypting messages directly
- B. Requires exactly the same amount of computation as symmetric encryption for bulk data
- C. Makes symmetric encryption obsolete, when in fact its overhead confines it mainly to signatures and key management
- D. Was invented decades after RSA, rather than being the concept RSA itself is built on

Correct Answer: C. Makes symmetric encryption obsolete, when in fact its overhead confines it mainly to signatures and key management

Explanation: Public-key cryptography is computationally slow, so symmetric encryption remains essential for bulk data encryption.

46. NIST's CTR_DRBG, the modern standard block-cipher-based PRNG, operates through which three phases?

- A. Initialize (seed from entropy), Generate (encrypt and increment a counter), and Update (periodically reseed with fresh entropy)
- B. Shuffle, Swap, and Output, exactly as in the RC4 algorithm
- C. Encrypt, Decrypt, and Re-encrypt, exactly as in Triple DES
- D. Select primes, Compute totient, and Generate keys, exactly as in RSA

Correct Answer: A. Initialize (seed from entropy), Generate (encrypt and increment a counter), and Update (periodically reseed with fresh entropy)

Explanation: NIST SP 800-90A CTR_DRBG defines instantiation (initialize), generate (produce pseudorandom output), and reseed/update.

47. For an elliptic curve defined over the real numbers as $y^2 = x^3 + ax + b$, adding two distinct points P and Q geometrically involves:

- A. Multiplying the x-coordinates of P and Q together to get the new x-coordinate
- B. Drawing the line through P and Q, finding its third intersection with the curve, and reflecting that point across the x-axis
- C. Reflecting P alone across the y-axis without ever considering Q's position
- D. Averaging the x-coordinates and y-coordinates of P and Q directly

Correct Answer: B. Drawing the line through P and Q, finding its third intersection with the curve, and reflecting that point across the x-axis

Explanation: The chord-and-tangent rule finds the third intersection point R of line PQ with the curve and reflects it across the x-axis.

48. The widely deployed stream cipher designed by Ron Rivest in 1987, later used in SSL/TLS and WEP, is called ____.

Correct Answer: RC4

Explanation: RC4 (Rivest Cipher 4) is the byte-oriented stream cipher designed by Ron Rivest.

49. Multiplication in $GF(2^m)$ is performed by:

- A. Multiplying the polynomials and then reducing the result modulo a fixed irreducible polynomial
- B. Simply XORing the two operands together, exactly as in addition
- C. Looking up the product directly in a table of prime factorizations
- D. Adding the two polynomials and discarding any bits beyond position m

Correct Answer: A. Multiplying the polynomials and then reducing the result modulo a fixed irreducible polynomial

Explanation: $GF(2^m)$ multiplication multiplies binary polynomials and reduces the result modulo an irreducible polynomial of degree m .

50. RC4 begins with a Key Scheduling Algorithm (KSA) that:

- A. Fixes the state array S to a constant, unchanging table for every key
- B. Directly encrypts the plaintext without ever touching a state array
- C. Uses the secret key to scramble a 256-byte state array S into a key-dependent permutation
- D. Generates a brand-new AES key for every byte of output produced

Correct Answer: C. Uses the secret key to scramble a 256-byte state array S into a key-dependent permutation

Explanation: The KSA initializes state $S[0..255]$ and performs 256 key-dependent swaps.

51. In evaluating candidate round functions F for a block cipher, the strict avalanche criterion (SAC) specifically requires that:

- A. Each output bit should change with probability one-half whenever a single input bit is inverted
- B. The round function should use exactly the same number of rounds as the key length in bits
- C. Every output bit should be a linear combination of the input bits
- D. The ciphertext should always be exactly the same length as the plaintext

Correct Answer: A. Each output bit should change with probability one-half whenever a single input bit is inverted

Explanation: SAC requires that inverting any single input bit flips each output bit with independent probability 0.5.

52. The Chinese Remainder Theorem allows a number to be uniquely reconstructed within a given range from:

- A. Its binary representation alone, without reference to any modulus
- B. A single large modulus formed by adding all the smaller moduli together
- C. Its remainders with respect to a set of moduli that share no common factors with one another
- D. Its prime factorization expressed in base two

Correct Answer: C. Its remainders with respect to a set of moduli that share no common factors with one another

Explanation: CRT guarantees a unique integer solution modulo the product of pairwise coprime moduli.

53. In RSA, the public key is the pair $\{e, n\}$ and the private key is the pair $\{____, n\}$.

Correct Answer: d

Explanation: The private decryption key pair in RSA is $\{d, n\}$ where $d = e^{-1} \bmod \phi(n)$.

54. A standard defense against RSA timing attacks is to:

- A. Ensure exponentiation always takes constant time, or add blinding by masking the ciphertext with a random factor before exponentiating
- B. Disable the use of the Chinese Remainder Theorem during decryption entirely
- C. Switch to a smaller RSA key size so that decryption completes faster
- D. Encrypt every message twice in succession using the same public key

Correct Answer: A. Ensure exponentiation always takes constant time, or add blinding by masking the ciphertext with a random factor before exponentiating

Explanation: Blinding multiplies ciphertext by $r^e \bmod n$ prior to decryption, removing data-dependent timing signatures.

55. Using the LCG recurrence $X_{n+1} = (5 \cdot X_n + 3) \bmod 16$ with seed $X_0 = 7$, the first two outputs X_1 and X_2 are:

- A. $X_1 = 1, X_2 = 6$
- B. $X_1 = 7, X_2 = 3$
- C. $X_1 = 6, X_2 = 1$
- D. $X_1 = 3, X_2 = 6$

Correct Answer: C. $X_1 = 6, X_2 = 1$

Explanation: $X_1 = (5 \cdot 7 + 3) \bmod 16 = 38 \bmod 16 = 6$; $X_2 = (5 \cdot 6 + 3) \bmod 16 = 33 \bmod 16 = 1$.

56. The security of the Blum Blum Shub generator rests on:

- A. The unpredictability of thermal noise sampled from physical hardware
- B. The difficulty of solving a system of linear congruential equations
- C. The length of the AES key used inside its internal encryption step
- D. The difficulty of factoring n back into its two large prime factors, the same hard problem underlying RSA

Correct Answer: D. The difficulty of factoring n back into its two large prime factors, the same hard problem underlying RSA

Explanation: BBS security reduces to the computational difficulty of calculating quadratic residues, equivalent to factoring $n = p \cdot q$.

57. Given $84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$, $\gcd(84, 126)$ equals:

- A. 21
- B. 42
- C. 84
- D. 126

Correct Answer: B. 42

Explanation: $\gcd(84, 126) = 2^1 \cdot 3^1 \cdot 7^1 = 42$.

58. Shannon's principle of diffusion is achieved mainly by:

- A. Replacing the round function with a simple table lookup
- B. Making the ciphertext statistics depend on the key in a highly complex way
- C. Reducing the key length needed to resist brute-force search
- D. Spreading the influence of each plaintext bit over many bits of the ciphertext

Correct Answer: D. Spreading the influence of each plaintext bit over many bits of the ciphertext

Explanation: Diffusion dissipates individual plaintext bit patterns across large regions of ciphertext.

59. Using the Chinese Remainder Theorem to split RSA decryption into two smaller modular exponentiations, one modulo p and one modulo q , speeds up decryption by roughly a factor of ____.

Correct Answer: 4

Explanation: CRT modular exponentiations are on half-sized moduli with cubic-complexity multiplication, yielding ~4x speedup.

60. In a Feistel cipher, the relationship that generates the right half of round i is:

- A. $R_i = F(L_{i-1}, K_i) \text{ XOR } R_{i-1} \text{ XOR } L_{i-1}$
- B. $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$
- C. $R_i = L_{i-1} \text{ AND } F(R_{i-1}, K_i)$
- D. $R_i = F(R_{i-1}, K_{i-1}) \text{ XOR } L_i$

Correct Answer: B. $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$

Explanation: In a Feistel round, $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$.

61. Given $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$, the Chinese Remainder Theorem gives the unique solution in the range 0 to 14 as:

- A. $x = 11$
- B. $x = 8$
- C. $x = 13$
- D. $x = 2$

Correct Answer: B. $x = 8$

Explanation: $8 \bmod 3 = 2$ and $8 \bmod 5 = 3$; 8 is the unique integer solution in $[0, 14]$.

62. The block cipher mode that turns a block cipher into a stream cipher by encrypting a counter value for each block and XORing it with the plaintext is called ____ mode

Correct Answer: Counter

Explanation: Counter (CTR) mode turns a block cipher into a stream cipher by encrypting sequential counter values.

63. The points on an elliptic curve, together with a defined addition rule and a point at infinity, form:

- A. An abelian group, satisfying closure, associativity, an identity element, inverses, and commutativity
- B. A Feistel structure, alternating substitution and permutation at each point
- C. A finite field in the same sense as $GF(p)$, with every point having a multiplicative inverse
- D. A simple ring with no guarantee that every point has an inverse

Correct Answer: A. An abelian group, satisfying closure, associativity, an identity element, inverses, and commutativity

Explanation: Under chord-and-tangent addition with point at infinity O , elliptic curve points form an abelian group.

64. A trap-door one-way function is defined as a function that is:

- A. A function that produces the exact same output regardless of its input
- B. Easy to invert for everyone, regardless of whether they hold any secret information
- C. Impossible to compute in both directions under any circumstances
- D. Easy to compute in one direction, and easy to invert only if you possess a secret piece of extra information

Correct Answer: D. Easy to compute in one direction, and easy to invert only if you possess a secret piece of extra information

Explanation: A trapdoor function is easy forward, hard to invert, but efficiently invertible with special trapdoor knowledge.

65. Which pair correctly matches an AES key size with its corresponding number of rounds?

- A. 192-bit key -> 10 rounds
- B. 256-bit key -> 12 rounds
- C. 192-bit key -> 12 rounds
- D. 128-bit key -> 14 rounds

Correct Answer: C. 192-bit key -> 12 rounds

Explanation: AES-128 uses 10 rounds, AES-192 uses 12 rounds, and AES-256 uses 14 rounds.

66. The avalanche effect, as illustrated by flipping a single bit of DES plaintext, refers to the property that:

- A. A small change in the key always leaves the ciphertext completely unchanged
- B. A small change in the input produces a large, seemingly unrelated change in the output ciphertext
- C. The ciphertext becomes shorter as more plaintext bits are altered
- D. Every round of DES must be repeated twice to detect tampering

Correct Answer: B. A small change in the input produces a large, seemingly unrelated change in the output ciphertext

Explanation: The avalanche effect ensures a 1-bit input flip alters approximately 50% of the ciphertext bits.

67. AES was selected by NIST through an open competition, with the winning algorithm originally named _____, designed by Daemen and Rijmen.

Correct Answer: Rijndael

Explanation: Rijndael, created by Vincent Rijmen and Joan Daemen, was standardized as AES.

68. Basic, unauthenticated Diffie-Hellman is vulnerable to a man-in-the-middle attack because:

- A. The discrete logarithm problem becomes trivially easy once two parties are involved
- B. The primitive root alpha must be kept secret, and any leak instantly reveals the key
- C. The protocol does not authenticate either party, allowing an attacker to establish separate shared keys with each side while relaying and altering messages
- D. Diffie-Hellman transmits the shared secret key directly, in plaintext, over the channel

Correct Answer: C. The protocol does not authenticate either party, allowing an attacker to establish separate shared keys with each side while relaying and altering messages

Explanation: Unauthenticated Diffie-Hellman does not verify sender identities, allowing active adversaries to intercept and negotiate split keys.

69. Double DES (encrypting twice with two different 56-bit keys) is considered insufficient mainly because it is vulnerable to:

- A. A chosen-plaintext attack that recovers the key from a single ciphertext block
- B. A simple brute-force search that is no harder than breaking single DES
- C. A meet-in-the-middle attack, which reduces its effective strength far below a true 112-bit key
- D. An attack that only works if ECB mode is used alongside it

Correct Answer: C. A meet-in-the-middle attack, which reduces its effective strength far below a true 112-bit key

Explanation: Meet-in-the-middle attacks find table collisions between forward and reverse encryptions in $\sim 2^{57}$ steps.

70. Inside a DES round, the 32-bit right half is expanded to 48 bits mainly so that it can be:

- A. Directly output as the new left half without further processing
- B. Combined via XOR with the 48-bit round subkey before passing through the S-boxes
- C. Compared bit-by-bit against the original plaintext for error checking
- D. Split evenly between the eight S-boxes without any subkey involved

Correct Answer: B. Combined via XOR with the 48-bit round subkey before passing through the S-boxes

Explanation: The E-box expands 32 bits to 48 bits to match the 48-bit subkey for XORing before substitution in the 8 S-boxes.

71. The block cipher mode in which each plaintext block is encrypted independently with the same key, and which reveals repeated patterns in the plaintext, is called ____ mode.

Correct Answer: Electronic-Codebook

Explanation: Electronic Codebook (ECB) encrypts each block in isolation, leaking statistical patterns.

72. If plaintext bit $P = 1$ is XORed with keystream bit $K = 0$, and the resulting ciphertext bit C is later XORed again with the same keystream bit K , the result is:

- A. 0, because two XOR operations always cancel out to zero
- B. 1, recovering the original plaintext bit exactly
- C. Undefined, since XOR cannot be reversed without also knowing P in advance
- D. 1, but only if K had instead been equal to 1

Correct Answer: B. 1, recovering the original plaintext bit exactly

Explanation: $C = 1 \text{ XOR } 0 = 1$; $C \text{ XOR } K = 1 \text{ XOR } 0 = 1$, restoring P since $P \text{ XOR } K \text{ XOR } K = P$.

73. Using the Chinese Remainder Theorem to speed up RSA's private-key operations works by:

- A. Splitting the decryption exponentiation into two smaller computations modulo p and modulo q , then recombining the results
- B. Replacing modular exponentiation entirely with simple XOR operations
- C. Avoiding the need to know either p or q once the public key has been published
- D. Reducing the RSA key size requirement from 2048 bits down to 512 bits

Correct Answer: A. Splitting the decryption exponentiation into two smaller computations modulo p and modulo q , then recombining the results

Explanation: RSA-CRT evaluates $C^{d \bmod p-1} \bmod p$ and $C^{d \bmod q-1} \bmod q$ separately, recombining via CRT.

74. After the KSA, RC4's Pseudo-Random Generation Algorithm (PRGA) produces the keystream by:

- A. Continuously swapping entries within the permuted state array S and combining selected entries to output one byte at a time
- B. Applying the SHA-1 hash function repeatedly to the previous keystream byte
- C. Reading the secret key directly, byte by byte, without any further processing
- D. Encrypting a running counter with a block cipher and outputting the ciphertext bytes

Correct Answer: A. Continuously swapping entries within the permuted state array S and combining selected entries to output one byte at a time

Explanation: PRGA updates pointer indices i and j , swaps $S[i]$ and $S[j]$, and outputs $S[(S[i]+S[j]) \bmod 256]$ per loop.

75. A network structure built from alternating layers of substitution and permutation across multiple rounds, as proposed by Shannon and used in ciphers such as DES, is called a substitution-_____ network.

Correct Answer: permutation

Explanation: A Substitution-Permutation Network (SPN) alternates S-boxes (confusion) and P-boxes (diffusion).

76. Which mode of operation is generally considered to have no safe use case for general-purpose encryption of typical application data?

- A. CTR
- B. ECB
- C. CBC
- D. OFB

Correct Answer: B. ECB

Explanation: ECB mode encrypts identical plaintext blocks to identical ciphertext blocks, failing semantic security.

77. Choosing a very small public exponent such as $e = 3$ without message padding creates a risk known as:

- A. A doubling of the ciphertext length compared to using a larger exponent
- B. An immediate leak of the private exponent d to any observer of the ciphertext
- C. A complete loss of the ability to decrypt the message even with the correct key
- D. A cube-root attack, where the same message sent to multiple recipients can be recovered without factoring n

Correct Answer: D. A cube-root attack, where the same message sent to multiple recipients can be recovered without factoring n

Explanation: Without padding, small exponent RSA ($e=3$) is vulnerable to Håstad's broadcast attack via CRT and direct root extraction.

78. When selecting a mode of operation for high-throughput, parallelizable, general-purpose encryption, the most appropriate recommendation is typically:

- A. ECB mode
- B. XTS-AES, regardless of whether the data is stored or transmitted
- C. CFB mode with a very small segment size
- D. CTR mode

Correct Answer: D. CTR mode

Explanation: CTR keystream generation depends solely on the block index counter, enabling full parallelization across CPU cores.

79. A mode of operation is best defined as:

- A. A set of rules describing how a block cipher should be applied to encrypt data longer than a single block
- B. An alternative block cipher algorithm used only for very large files
- C. A method for generating the encryption key from a user-supplied password
- D. A hardware-only technique that cannot be implemented in software

Correct Answer: A. A set of rules describing how a block cipher should be applied to encrypt data longer than a single block

Explanation: A mode of operation defines algorithms and feedback chaining to securely encrypt sequences of arbitrary length.

80. Which requirement is essential when parameterizing an RSA-based PRNG such as Micali-Schnorr?

- A. e must be chosen equal to the private exponent d used for RSA decryption
- B. $\gcd(e, \phi(n)) = 1$, mirroring the same requirement used for ordinary RSA key generation
- C. The modulus n must be chosen to be a small prime number, unlike ordinary RSA
- D. The output length r must always exceed the modulus n 's bit length

Correct Answer: B. $\gcd(e, \phi(n)) = 1$, mirroring the same requirement used for ordinary RSA key generation

Explanation: For $x \mapsto x^e \bmod n$ to be an invertible permutation over $\mathbb{Z}_n^*$, e must be coprime to $\phi(n)$.

81. Given public parameters $q = 353$ and $\alpha = 3$, with Alice's private key $X_A = 97$ and public key $Y_A = 40$, and Bob's private key $X_B = 233$ and public key $Y_B = 248$, Alice computes the shared secret as:

- A. $K = X_A * X_B \bmod q = 97 * 233 \bmod 353$
- B. $K = Y_A^{X_B} \bmod q = 40^{233} \bmod 353$
- C. $K = Y_B^{X_A} \bmod q = 248^{97} \bmod 353$
- D. $K = \alpha^{(X_A + X_B)} \bmod q$, computed without ever using Y_A or Y_B

Correct Answer: C. $K = Y_B^{X_A} \bmod q = 248^{97} \bmod 353$

Explanation: In DH, Alice computes the shared secret as $K = Y_B^{X_A} \bmod q = 248^{97} \bmod 353$.

82. In Cipher Block Chaining (CBC) mode, the input to the encryption function for a given block is:

- A. The XOR of the current plaintext block with the next plaintext block
- B. The previous plaintext block encrypted a second time
- C. The XOR of the current plaintext block with the previous ciphertext block (or the IV, for the first block)
- D. The plaintext block alone, exactly as in ECB mode

Correct Answer: C. The XOR of the current plaintext block with the previous ciphertext block (or the IV, for the first block)

Explanation: In CBC mode, $\text{Input}_i = P_i \oplus C_{i-1}$ (with $C_0 = \text{IV}$).

83. The AES key expansion algorithm's $g()$ function, applied to certain words, involves:

- A. Rotating the bytes of the word, applying the S-box to each byte, then XORing in a round constant
- B. Reversing the entire 128-bit key and discarding half of it
- C. Multiplying the word by the AES modulus polynomial directly
- D. Applying the inverse ShiftRows transformation to the word

Correct Answer: A. Rotating the bytes of the word, applying the S-box to each byte, then XORing in a round constant

Explanation: The AES key schedule $g(w)$ routine applies RotWord (cyclic shift), SubWord (S-box), and XOR with Rcon.

84. Which of the following was NOT one of the finalist algorithms in the AES selection process?

- A. Twofish
- B. MARS
- C. Blowfish
- D. Serpent

Correct Answer: C. Blowfish

Explanation: Blowfish is a 64-bit cipher designed in 1993. The 5 AES finalists were Rijndael, Serpent, Twofish, MARS, and RC6.

85. NIST's selection of Rijndael as AES was notable because it was:

- A. Chosen without any competing candidate algorithms being considered
- B. The result of an open, multi-round public competition rather than an algorithm designed internally by the government
- C. The first cipher ever to be patented by a government standards body
- D. Selected purely on the basis of being the fastest candidate in hardware

Correct Answer: B. The result of an open, multi-round public competition rather than an algorithm designed internally by the government

Explanation: NIST hosted an open multi-round international competition with transparent public cryptanalysis.

86. In the Micali-Schnorr RSA-based PRNG, each stage encrypts the current internal state and then splits the result so that:

- A. Some bits feed back to seed the next stage while the remaining bits form part of the pseudorandom output
- B. All of the output bits are discarded and only the feedback bits are ever kept
- C. Half of the RSA modulus n is regenerated fresh at every single stage
- D. The entire output becomes the feedback value for the next stage, with nothing output

Correct Answer: A. Some bits feed back to seed the next stage while the remaining bits form part of the pseudorandom output

Explanation: Micali-Schnorr splits $x_i^e \bmod n$ into pseudorandom output bits and seed feedback for the next stage.

87. Increasing the block size of a Feistel cipher generally:

- A. Improves security through greater diffusion, at the cost of more processing time per block
- B. Has no measurable effect on either security or processing time
- C. Removes the need for multiple rounds of processing
- D. Weakens security because larger blocks leak more statistical structure

Correct Answer: A. Improves security through greater diffusion, at the cost of more processing time per block

Explanation: Larger blocks improve diffusion and resist birthday/dictionary attacks, but increase register size and compute cycles.

88. Triple DES (3DES), typically applying Encrypt-Decrypt-Encrypt with two or three keys, achieves an effective key length of up to:

- A. 168 bits, using three independent keys
- B. 56 bits, the same as single DES
- C. 224 bits, matching twice the length of AES-256's key
- D. 112 bits, even when three independent keys are used

Correct Answer: A. 168 bits, using three independent keys

Explanation: 3DES using 3 independent 56-bit DES keys has a nominal total key length of $3 * 56 = 168$ bits.

89. The Diffie-Hellman key exchange allows two parties to establish a shared secret key by:

- A. Directly transmitting the shared secret key encrypted under RSA
- B. Meeting in person beforehand to physically exchange the key on paper
- C. Using a trusted third party to generate and distribute the key to both sides
- D. Exchanging public values computed from private exponents over an insecure channel, without ever transmitting the secret key itself

Correct Answer: D. Exchanging public values computed from private exponents over an insecure channel, without ever transmitting the secret key itself

Explanation: Both sides exchange only modular exponentiations $\alpha^X \bmod q$, computing the shared secret independently.

90. A common practical design pairs a TRNG with a PRNG by:

- A. Alternating output bits strictly one-for-one between the TRNG and the PRNG
- B. Using the PRNG to produce entropy that is then fed back into the TRNG's hardware
- C. Using the TRNG's physical entropy to seed the PRNG, combining true unpredictability with the PRNG's speed and reproducibility
- D. Running both generators completely independently and never combining their outputs

Correct Answer: C. Using the TRNG's physical entropy to seed the PRNG, combining true unpredictability with the PRNG's speed and reproducibility

Explanation: A physical TRNG harvests hardware entropy to seed/reseed a high-throughput deterministic PRNG.

91. The Prime Number Theorem indicates that, near a large integer n , prime numbers occur on average about once every:

- A. 2 integers checked, since every other integer is prime past a certain point
- B. $\ln(n)$ integers checked
- C. n integers checked, regardless of how large n becomes
- D. $\sqrt{n}$ integers checked, matching the trial-division bound

Correct Answer: B. $\ln(n)$ integers checked

Explanation: The Prime Number Theorem establishes that prime density near integer n is asymptotically $1 / \ln(n)$.

92. The Intel Digital Random Number Generator (DRNG), used in Intel processors since 2012, combines which stages in its pipeline?

- A. A thermal-noise entropy source, a bias-removing conditioner, and a CTR_DRBG stage exposed through the RDRAND instruction
- B. A user-supplied password hashed once with SHA-1 and output directly
- C. Two independent TRNGs whose raw, unconditioned outputs are simply concatenated
- D. An LCG seeded from the system clock, followed directly by RC4 encryption

Correct Answer: A. A thermal-noise entropy source, a bias-removing conditioner, and a CTR_DRBG stage exposed through the RDRAND instruction

Explanation: Intel DRNG features a thermal noise entropy source latch, CBC-MAC conditioner, and AES-CTR DRBG engine.

93. Which AES round transformation is the only one that is non-linear, providing the algorithm's confusion?

- A. SubBytes
- B. ShiftRows
- C. AddRoundKey
- D. MixColumns

Correct Answer: A. SubBytes

Explanation: SubBytes uses inversion in $GF(2^8)$ to provide the cipher's essential non-linear confusion.

94. Testing whether 29 is prime by checking divisibility only up to $\sqrt{29} \approx 5.4$ works because:

- A. Any factor larger than the square root of a number must be paired with a factor smaller than the square root
- B. Numbers greater than the square root are automatically assumed to be prime
- C. The square root always equals exactly half of the number being tested
- D. 29 is a special case, and this shortcut does not generalize to any other number

Correct Answer: A. Any factor larger than the square root of a number must be paired with a factor smaller than the square root

Explanation: If $n = a * b$, both factors cannot exceed $\sqrt{n}$; at least one factor must be $\leq \sqrt{n}$.

95. A mode of operation that allows random access to any individual block without first decrypting all preceding blocks is best exemplified by:

- A. ECB mode used together with an initialization vector
- B. CTR mode, since each block's keystream depends only on its own counter value
- C. CFB mode, since it processes data in variable-size segments
- D. CBC mode, since each block depends on the ciphertext immediately before it

Correct Answer: B. CTR mode, since each block's keystream depends only on its own counter value

Explanation: CTR keystream blocks are computed directly from block counter indices, allowing instant random access.

96. A defining advantage of the Feistel structure is that:

- A. No subkey schedule is required, since the same key is reused every round
- B. Decryption uses the identical algorithm as encryption, only with the subkeys applied in reverse order
- C. The round function F must itself be a reversible, invertible operation
- D. Encryption and decryption each require a completely separate algorithm

Correct Answer: B. Decryption uses the identical algorithm as encryption, only with the subkeys applied in reverse order

Explanation: Feistel networks decrypt using the identical forward circuit simply by reversing the subkey schedule.

97. In Diffie-Hellman, both public parameters q (a prime) and α (a primitive root of q) are:

- A. Derived mathematically from each party's private exponent after the exchange
- B. Generated fresh for every single message exchanged between the parties
- C. Kept secret and known only to the two communicating parties
- D. Known publicly to everyone, including any potential attacker

Correct Answer: D. Known publicly to everyone, including any potential attacker

Explanation: Modulus q and generator α are public parameters known to everyone.

98. A True Random Number Generator (TRNG) differs from a PRNG chiefly in that a TRNG:

- A. Draws its randomness from an unpredictable physical process, such as thermal noise, rather than a deterministic algorithm
- B. Cannot be used to seed any other type of random number generator
- C. Requires a secret key to be shared in advance between sender and receiver
- D. Always produces output faster than any PRNG can achieve

Correct Answer: A. Draws its randomness from an unpredictable physical process, such as thermal noise, rather than a deterministic algorithm

Explanation: TRNGs harvest nondeterministic physical noise, whereas PRNGs apply deterministic mathematical algorithms to expand seeds.

99. An ideal block cipher mapping n bits to n bits allows how many possible reversible mappings?

- A. 2^n
- B. $n!$
- C. $(2^n)!$
- D. $2^{(2n)}$

Correct Answer: C. $(2^n)!$

Explanation: There are 2^n possible input states, and any reversible mapping is a permutation of these states, yielding $(2^n)!$ possible bijections.

100. The primary advantage of asymmetric key cryptography over symmetric key cryptography in large communication networks is that:

- A. Asymmetric cryptography is computationally faster for bulk file transfers
- B. It scales with N key pairs for N users instead of $N*(N-1)/2$ shared secret keys
- C. It uses shorter keys to achieve the same bit-strength security as AES
- D. It eliminates the need for mathematical modular arithmetic

Correct Answer: B. It scales with N key pairs for N users instead of $N*(N-1)/2$ shared secret keys

Explanation: In a network of N users, symmetric encryption requires $N*(N-1)/2$ pre-shared secret keys, whereas public-key cryptography requires only N key pairs.